

Harshit Mittal

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Experience

SatEx Systems

Aug 2026 – Oct 2026

AI Engineering and Consulting Intern

United Kingdom

- Authored a research brief evaluating various propagation strategies and specified a fidelity-agnostic propagator interface, enabling quick analytical previews to run concurrently with high-fidelity numerical refinement without coupling downstream modules.
- Designed a layered AI/ML optimization pipeline to reduce expensive high-fidelity evaluations with a C4-model software architecture (Context, Container, Component, Code) guaranteeing UI responsiveness during background computation.

USIC&T

Jan 2025 – May 2025

AI Research Intern

Delhi, India

- Designed and evaluated novel piecewise activation functions for deep learning models, benchmarked across Iris, Breast Cancer Wisconsin, MNIST, and CIFAR-10 datasets using CNN and VGG16 architectures.
- Reduced information loss in data and improved classification accuracy through activation function optimization and class-balancing techniques, with results validated with Springer and IEEE reviewers.

CFEES, DRDO

Aug 2023 – Oct 2023

Project Trainee

Delhi, India

- Engineered and deployed a peer-reviewed 3D facial recognition pipeline that upgraded CCTV footage to point-clouds, directly improving the department's active production security posture.
- Optimized cross-platform Flutter application performance to boost end-user satisfaction, establishing rigid code standards through meticulous Git history and documentation tracking.

Education

University of Leeds

September 2025 - September 2026

Master of Science in Advanced Computer Science (First Class)

Leeds, United Kingdom

- **Thesis** [arXiv:2607.27835]: The design and implementation of SAFViT, a novel hierarchical encoder-decoder architecture integrating Spatial Adaptive Fusion over the sliding window backbone, outperforming models like CellViT across minority dead cells on the PanNuke dataset.

Maharaja Agrasen Institute of Technology

Sept 2021 - August 2025

Bachelors of Technology in Computer Science and Engineering (First Class)

Delhi, India

Projects

HRGoggle: Multimodal AI Mock Interview Platform | [JavaScript](#), [RAG](#), [AI](#), [Computer Vision](#), [NLP](#) | [WebApp](#)

- Built a fully client-side AI interview platform implementing a lightweight retrieval-augmented generation (RAG) pipeline, parsing resumes, retrieving skill-relevant context, and using it to generate personalized, resume-grounded interview questions.
- Integrated real-time head-pose and eye-contact tracking during live webcam sessions and real-time speech-analytics pipeline to compute words-per-minute, filler-word frequency, and per-question response time, aggregating scores into interactive visualizations for session-level performance comparison.

MealWeDeal: On-Device Multimodal Nutrition Assistant | [JavaScript](#), [Transformers](#), [RAG](#), [Computer Vision](#) | [WebApp](#)

- Built a fully client-side nutrition web app performing computer vision, retrieval-augmented generation (RAG), and macro calculation with a CLIP zero-shot image classification model to identify meals from photos.
- Engineered a RAG retrieval pipeline using sentence embeddings and cosine similarity, combined with a custom macro-alignment scoring algorithm to rank food swaps by how closely their calorie, protein, and fat proportions matched user-defined numeric targets with a scaling engine that proportionally recalculates calories, protein, carbs, and fat from a baseline serving size to a user-entered gram weight.

Skills

Technological Skills

Python, C, C++, SQL, Artificial Intelligence (AI), Retrieval-Augmented Generation (RAG), Docker and Azure ML.

Soft Skills

Collaboration, Quick Learning, Research, Communication and Planning.